



Lecture 3.2: Sag and Other Topics in Overhead Transmission

EEE 321: Power System I

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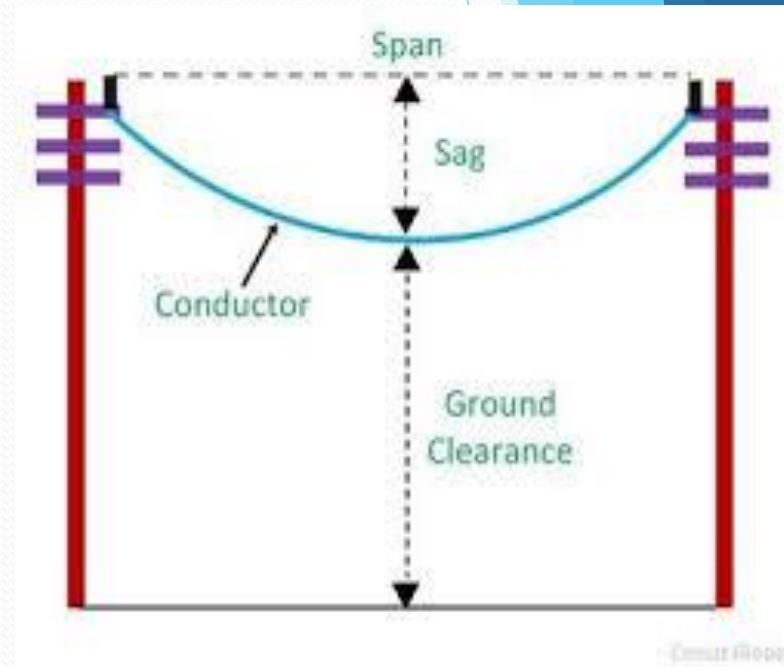
Presentation Outline

- ❑ Sag in Overhead Transmission Lines
- ❑ Factors affecting Sag
- ❑ Sag Calculation
- ❑ Examples
- ❑ Ferranti Effect
- ❑ Surge Impedance Loading (SIL)

Sag in Overhead Transmission Lines

While erecting an overhead line, it is very important that conductors are under safe tension. If the conductors are too much stretched between supports in a bid to save conductor material, the stress in the conductor may reach unsafe value and in certain cases the conductor may break due to excessive tension. In order to permit safe tension in the conductors, they are not fully stretched but are allowed to have a dip or sag.

The difference in level between points of supports and the lowest point on the conductor is called sag.



Sag in Overhead Transmission Lines

Factors Affecting The Sag:

1. Conductor Weight:

The weight of the conductor, including any ice or snow accumulation, directly affects sag. Heavier conductors experience greater sag i.e. the **sag of an overhead line is directly proportional to the conductor weight.**

2. Span Length:

Longer spans between supports result in increased sag. **The sag is directly proportional to the square of the span length.**

3. Tension:

Tension in the conductor is inversely proportional to sag. Higher tension reduces sag, while lower tension increases it. Tension is influenced by factors like wind, ice loading, and conductor weight.

4. Temperature:

Temperature changes cause the conductor to expand or contract. **Higher temperatures increase sag as the conductor expands, while lower temperatures reduce sag as it contracts.**

5. Environmental Factors:

Wind: **Wind exerts a force on the conductor, increasing sag in the direction of the wind.** Wind can also cause the conductor to sway, further increasing sag.

Ice Loading: **Ice or snow accumulation on the conductor adds weight, increasing sag.** Ice loading can also affect tension and increase the risk of conductor breakage.

Sag in Overhead Transmission Lines

Sag Calculation:

In an overhead line, the sag should be so adjusted that tension in the conductors is within safe limits. The tension is governed by conductor weight, effects of wind, ice loading and temperature variations. It is a standard practice to keep conductor tension less than 50% of its ultimate tensile strength *i.e.*, minimum factor of safety in respect of conductor tension should be 2. We shall now calculate sag and tension of a conductor when (i) supports are at equal levels and (ii) supports are at unequal levels.

i. When supports are at equal levels:

(i) When supports are at equal levels. Consider a conductor between two equilevel supports *A* and *B* with *O* as the lowest point as shown in Fig. 8.24. It can be proved that lowest point will be at the mid-span.

Let

l = Length of span

w = Weight per unit length of conductor

T = Tension in the conductor.

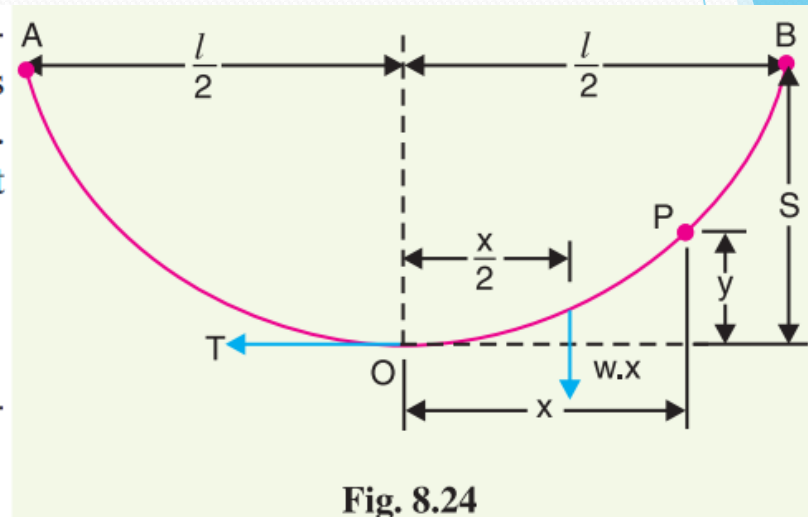


Fig. 8.24

Consider a point *P* on the conductor. Taking the lowest point *O* as the origin, let the co-ordinates of point *P* be x and y . Assuming that the curvature is so small that curved length is equal to its horizontal projection (*i.e.*, $OP = x$), the two forces acting on the portion *OP* of the conductor are :

- (a) The weight $w x$ of conductor acting at a distance $x/2$ from *O*.
- (b) The tension T acting at *O*.

Sag in Overhead Transmission Lines

Sag Calculation:

i. When supports are at equal levels:

Equating the moments of above two forces about point O , we get,

$$T y = w x \times \frac{x}{2}$$

or

$$y = \frac{w x^2}{2 T}$$

The maximum dip (sag) is represented by the value of y at either of the supports A and B .

At support A , $x = l/2$ and $y = S$

$\therefore$

$$\text{Sag, } S = \frac{w(l/2)^2}{2T} = \frac{w l^2}{8 T}$$

Sag in Overhead Transmission Lines

SAG CALCULATION:

ii. When supports are at unequal levels:

(ii) **When supports are at unequal levels.** In hilly areas, we generally come across conductors suspended between supports at unequal levels. Fig. 8.25 shows a conductor suspended between two supports A and B which are at different levels. The lowest point on the conductor is O .

Let

l = Span length

h = Difference in levels between two supports

x_1 = Distance of support at lower level (*i.e.*, A) from O

x_2 = Distance of support at higher level (*i.e.*, B) from O

T = Tension in the conductor

If w is the weight per unit length of the conductor, then,

$$\text{Sag } S_1 = \frac{w x_1^2}{2T}$$

and
$$\text{Sag } S_2 = \frac{w x_2^2}{2T}$$

Also
$$x_1 + x_2 = l$$

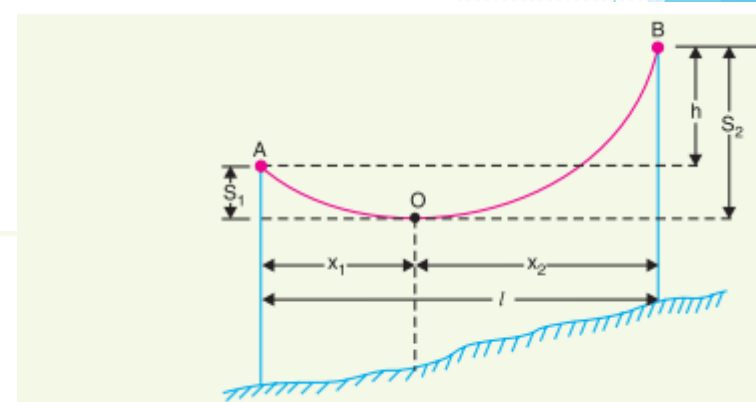


Fig. 8.25

...(i)

$$* y = \frac{w x^2}{2T}$$

At support A , $x = x_1$ and $y = S_1$.

$$\therefore S_1 = \frac{w x_1^2}{2T}$$

Sag in Overhead Transmission Lines

SAG CALCULATION:

ii. When supports are at unequal levels:

Now
$$S_2 - S_1 = \frac{w}{2T} [x_2^2 - x_1^2] = \frac{w}{2T} (x_2 + x_1)(x_2 - x_1)$$

$\therefore$
$$S_2 - S_1 = \frac{w l}{2T} (x_2 - x_1) \quad [\because x_1 + x_2 = l]$$

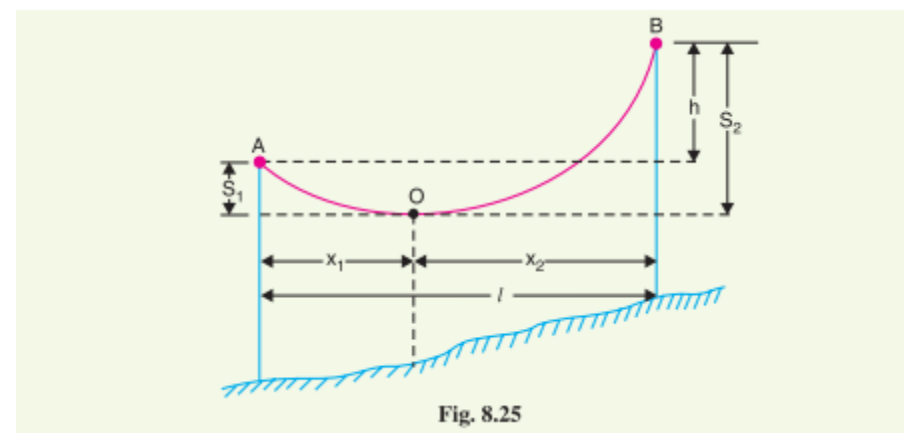
But
$$S_2 - S_1 = h$$

$\therefore$
$$h = \frac{w l}{2T} (x_2 - x_1)$$

or
$$x_2 - x_1 = \frac{2 T h}{w l}$$

Solving exps. (i) and (ii), we get,

$$x_1 = \frac{l}{2} - \frac{T h}{w l}$$
$$x_2 = \frac{l}{2} + \frac{T h}{w l}$$



Having found x_1 and x_2 , values of S_1 and S_2 can be easily calculated.

Sag in Overhead Transmission Lines

SAG CALCULATION:

Effects of wind and ice loading:

Effect of wind and ice loading. The above formulae for sag are true only in still air and at normal temperature when the conductor is acted by its weight only. However, in actual practice, a conductor may have ice coating and simultaneously subjected to wind pressure. The weight of ice acts vertically downwards *i.e.*, in the same direction as the weight of conductor. The force due to the wind is assumed to act horizontally *i.e.*, at right angle to the projected surface of the conductor. Hence, the total force on the conductor is the vector sum of horizontal and vertical forces as shown in Fig. 8.26 (iii).

Total weight of conductor per unit length is

$$w_t = \sqrt{(w + w_i)^2 + (w_w)^2}$$

where

w = weight of conductor per unit length

= conductor material density $\times$ volume per unit length

w_i = weight of ice per unit length

= density of ice $\times$ volume of ice per unit length

= density of ice $\times \frac{\pi}{4} [(d + 2t)^2 - d^2] \times 1$

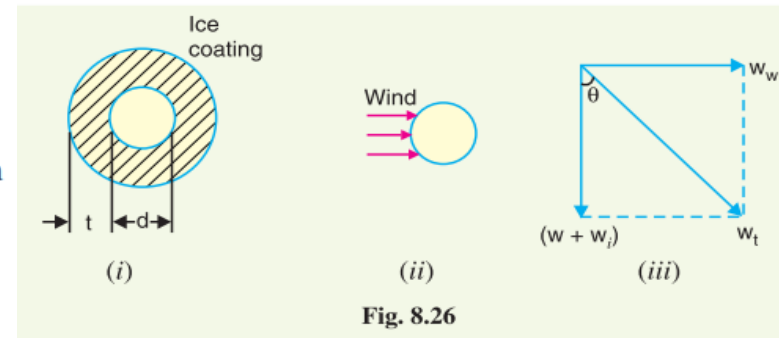
= density of ice $\times \pi t (d + t)^*$

w_w = wind force per unit length

= wind pressure per unit area $\times$ projected area per unit length

= wind pressure $\times [(d + 2t) \times 1]$

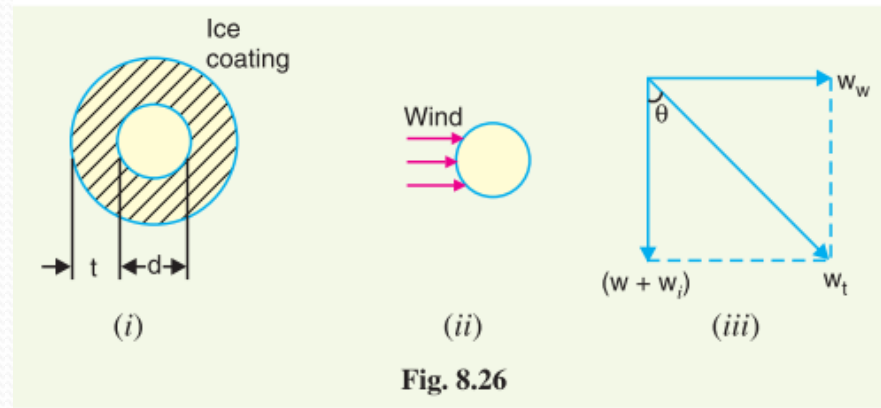
$$\text{Volume of ice per unit length} = \frac{\pi}{4} [(d + t)^2 - d^2] \times 1 = \frac{\pi}{4} [4dt + 4t^2] = \pi t (d + t)$$



Sag in Overhead Transmission Lines

SAG CALCULATION:

Effects of wind and ice loading:



When the conductor has wind and ice loading also, the following points may be noted :

- (i) The conductor sets itself in a plane at an angle θ to the vertical where

$$\tan \theta = \frac{w_w}{w + w_i}$$

- (ii) The sag in the conductor is given by :

$$S = \frac{w_t l^2}{2T}$$

Hence S represents the slant sag in a direction making an angle θ to the vertical. *If no specific mention is made in the problem, then slant sag is calculated by using the above formula.*

- (iii) The vertical sag = $S \cos \theta$

Sag in Overhead Transmission Lines

EXAMPLES:

Example 8.17. A 132 kV transmission line has the following data :

Wt. of conductor = 680 kg/km ; Length of span = 260 m

Ultimate strength = 3100 kg ; Safety factor = 2

Calculate the height above ground at which the conductor should be supported. Ground clearance required is 10 metres.

Solution.

Wt. of conductor/metre run, $w = 680/1000 = 0.68$ kg

Working tension, $T = \frac{\text{Ultimate strength}}{\text{Safety factor}} = \frac{3100}{2} = 1550$ kg

Span length, $l = 260$ m

$\therefore$ Sag = $\frac{w l^2}{8T} = \frac{0.68 \times (260)^2}{8 \times 1550} = 3.7$ m

$\therefore$ Conductor should be supported at a height of $10 + 3.7 = 13.7$ m

Sag in Overhead Transmission Lines

EXAMPLES:

Example 8.18. A transmission line has a span of 150 m between level supports. The conductor has a cross-sectional area of 2 cm^2 . The tension in the conductor is 2000 kg. If the specific gravity of the conductor material is 9.9 gm/cm^3 and wind pressure is 1.5 kg/m length, calculate the sag. What is the vertical sag?

Solution.

Span length, $l = 150 \text{ m}$; Working tension, $T = 2000 \text{ kg}$

Wind force/m length of conductor, $w_w = 1.5 \text{ kg}$

Wt. of conductor/m length, $w = \text{Sp. Gravity} \times \text{Volume of 1 m conductor}$
 $= 9.9 \times 2 \times 100 = 1980 \text{ gm} = 1.98 \text{ kg}$

Total weight of 1 m length of conductor is

$$w_t = \sqrt{w^2 + w_w^2} = \sqrt{(1.98)^2 + (1.5)^2} = 2.48 \text{ kg}$$

$$\therefore \text{Sag, } S = \frac{w_t l^2}{8T} = \frac{2.48 \times (150)^2}{8 \times 2000} = 3.48 \text{ m}$$

This is the value of slant sag in a direction making an angle θ with the vertical. Referring to Fig. 8.27, the value of θ is given by ;

$$\tan \theta = w_w/w = 1.5/1.98 = 0.76$$

$$\therefore \theta = \tan^{-1} 0.76 = 37.23^\circ$$

$$\therefore \text{Vertical sag} = S \cos \theta$$
$$= 3.48 \times \cos 37.23^\circ = 2.77 \text{ m}$$

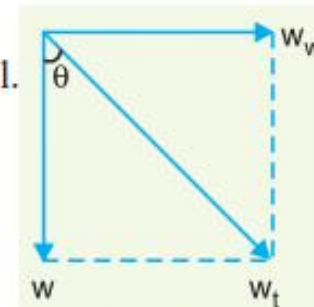


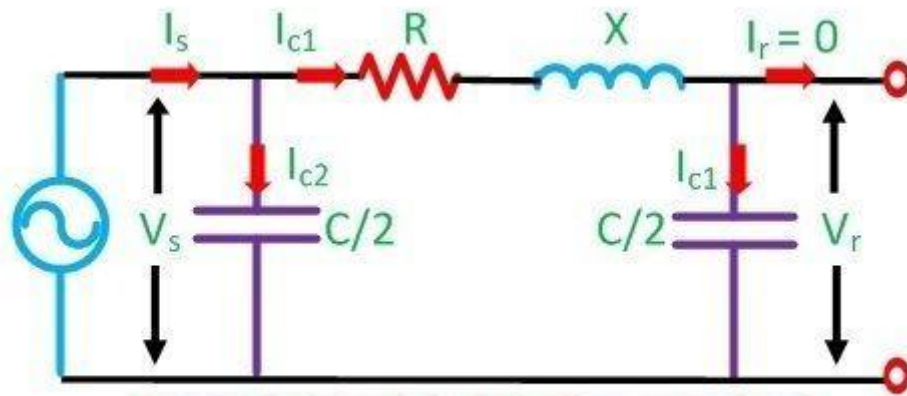
Fig. 8.27

Ferranti Effect

The effect in which the voltage at the receiving end of the transmission line is more than the sending voltage is known as the Ferranti effect. Such type of effect mainly occurs because of light load or open circuit at the receiving end.

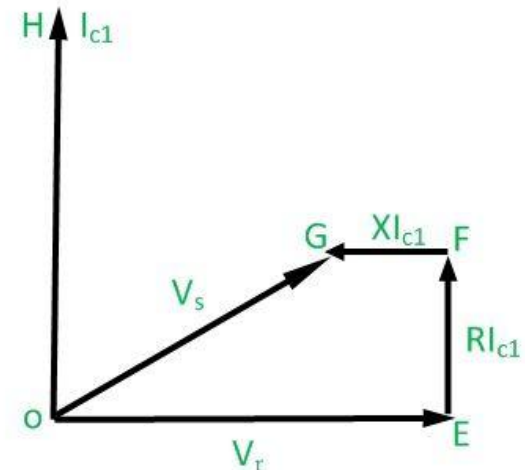
Detail explanation of the Ferranti effect by considering a nominal pi (π) model:

Let us consider the long transmission line in which OE represents the receiving end voltage; OH represent the current through the capacitor at the receiving end. The phasor FE represents the voltage drop across the resistance R. The voltage drop across the X (inductance). The phasor OG represents the sending end voltage under a no-load condition.



Nominal pi model of the line at no load.

Circuit Globe



Phasor Diagram

Circuit Globe

It is seen from phasor diagram that $OE > OG$. In other words, the voltage at the receiving end is greater than the voltage at the sending end when the line is at no load.

Ferranti Effect

How to reduce Ferranti effect:

Electrical devices are designed to work at some particular voltage. If the voltages are high at the user ends their equipment get damaged, and their windings burn because of high voltage. Ferranti effect on long transmission lines at low load or no load increases the receiving end voltage.

1. Shunt Reactor

This voltage can be controlled by placing the shunt reactors at the receiving end of the lines. Shunt reactor is an inductive current element connected between line and neutral to compensate the capacitive current from transmission lines. When this effect occurs in long transmission lines, shunt reactors compensate the capacitive VAR of the lines and therefore the voltage is regulated within the prescribed limits.

2. Surge Impedance Loading (SIL):

SIL is the load level at which the reactive power generated by the line capacitance is equal to the reactive power absorbed by the line inductance. Operating a transmission line at or near its SIL minimizes the Ferranti effect because it balances the capacitive and inductive effects. While achieving exact SIL operation may not always be feasible, operating closer to it can significantly reduce the voltage rise.

3. FACTS Devices:

Flexible Alternating Current Transmission Systems (FACTS) devices, such as Thyristor Controlled Reactors (TCRs), can provide dynamic reactive power compensation. TCRs offer a more flexible and controllable solution compared to fixed shunt reactors. By adjusting the reactive power supplied or absorbed by the TCR, the voltage profile can be effectively managed, even under varying load conditions. Other FACTS devices like STATCOMs, dynamic voltage restorers, and UPFCs can also be used for reactive power compensation and Ferranti effect mitigation.

Surge Impedance Loading (SIL)

Surge Impedance is the **characteristic impedance** of a lossless Transmission Line. As it is not involved with the load impedance, it is also called the **Natural Impedance**. When the line is assumed to be lossless, it means that the series resistance and shunt conductance is negligible i.e zero for power lines.

Therefore, we can say:

Shunt Conductance (G) = 0 and Series Resistance (R) = 0

Capacitive VAR = Inductive VAR

$$\frac{V^2}{X_c} = I^2 X_L$$

Upon simplifying Characteristic Impedance of a line can be represented as:

$$Z_c = \sqrt{\frac{L}{C}}$$

Z_c is the surge impedance. We can derive the surge impedance formula in the following way as well.

$$Z_s = \sqrt{\frac{z}{y}}$$

where 'z' is the series impedance per unit length per phase and 'y' is the shunt admittance per unit length per phase.

Where,

V = Phase voltage

I = Line Current

X_c = Capacitive reactance /phase

X_L = Inductive reactance / phase

Surge Impedance Loading (SIL)

Surge Impedance Loading is the loading on the line whenever the load impedance matches exactly with that of the surge impedance of the transmission line. The surge impedance loading can also be referred to as the Characteristic Impedance Loading or the ideal loading or the ideal power transfer capability. Therefore, at SIL the load impedance is equal to the characteristic impedance.

If P_0 is its natural load of the lines, $(SIL)_{1\phi}$ of the line per phase **Surge impedance loading formula** of single-phase circuit in watt is;

$$(SIL)_{1\phi} = P_0 = V_p I_p \cos\phi$$

$\cos\phi = 1$ as load is resistive for Z_s

$$(SIL)_{1\phi} = \frac{V_p V_p}{Z_s} = \frac{V_p^2}{Z_s}$$

Surge impedance loading formula of three-phase circuit in watt,

$$(SIL)_{3\phi} = 3P_0 = 3V_p I_p \cos\phi$$

$\cos\phi = 1$ as load is resistive for Z_s

$$(SIL)_{3\phi} = \frac{3V_p^2}{Z_s}$$

Surge Impedance Loading (SIL)

Example: 3 Phase 400 KV, 50 Hz, 300 km long lossless transmission line has a series impedance $Z = 0.3 \Omega / \text{km}$ and a shunt admittance of $Y = 3.75 \times 10^{-6} \text{ S} / \text{km}$.

Find

- a) Surge impedance
- b) Propagation constant
- c) Surge impedance loading(SIL)

Solution:

Given,

3 phase, 400 KV, 50 Hz Line

Length = 300 km (lossless line)

$Z = 0.3 \Omega / \text{km}$

$Y = 3.75 \times 10^{-6} \text{ S} / \text{km}$

b) Propagation constant

$$\gamma = \sqrt{ZY}$$

$$\gamma = \sqrt{0.3 \times 3.75 \times 10^{-6}}$$

$$\gamma = 0.00106$$

$$\gamma = 1.06 \times 10^{-3} \text{ rad/km}$$

a) Surge impedance

$$Z_s = \sqrt{\frac{Z}{Y}}$$

$$Z_s = \sqrt{\frac{0.3}{3.75 \times 10^{-6}}}$$

$$Z_s = 282.84 \Omega$$

c) Surge impedance loading(SIL)

$$\text{SIL} = \frac{V_R^2}{Z_c}$$

$$\text{SIL} = \frac{(400 \times 10^3)^2}{282.84}$$

$$\text{SIL} = 565.69 \text{ MW}$$

Reference: Principles of Power System, V.K. Mehta and Rohit Mehta, S. Chand & Company.



QUESTIONS AND DISCUSSIONS

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